

QWMO: A Quantum Wave-function Inspired Metaheuristic for Multimodal Optimization

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Abstract

This paper introduces QWMO (Quantum Wave-function Metaheuristic Optimizer), a quantum-inspired population-based optimization framework designed to investigate the role of probabilistic operators in balancing exploration and exploitation in multimodal optimization landscapes.

The proposed framework combines three operators: (i) Adaptive Orbital Sampling, which controls Gaussian search dispersion according to relative solution quality; (ii) Pauli-Inspired Exclusion, which preserves population diversity through orthogonal displacement dynamics; and (iii) Adaptive Quantum Escape, which enables stagnating agents to probabilistically leave local optima through stochastic relocation.

Unlike classical physics-inspired optimizers relying on deterministic force interactions, QWMO models search dynamics through wave-function-guided stochastic transitions. Experiments on five representative CEC-style benchmark functions in 30 dimensions with 30 independent runs indicate that QWMO consistently outperforms its direct physics-inspired counterparts ASO and AOS under Wilcoxon signed-rank analysis ($p < 0.05$), while maintaining competitive behavior against classical swarm-based optimizers on multimodal and hybrid landscapes.

An ablation study further shows that QWMO's behavior emerges from the interaction between adaptive orbital sampling, diversity-preserving exclusion, and stochastic escape dynamics, rather than from any single operator alone.

Keywords: metaheuristic optimization, quantum-inspired optimization, multimodal optimization, swarm intelligence, stochastic optimization, Markov process

1. Introduction

Metaheuristic algorithms have become one of the dominant paradigms for black-box global optimization. Population-based frameworks such as Particle Swarm Optimization (PSO) [1], Genetic Algorithms (GA) [2], Grey Wolf Optimizer (GWO) [3], and Whale Optimization Algorithm (WOA) [4] have been widely applied to continuous, discrete, and hybrid optimization problems.

Despite their empirical success, many metaheuristic optimizers still suffer from premature convergence, loss of diversity, and unstable exploration–exploitation balance, particularly in highly multimodal landscapes. This limitation becomes more pronounced in high-dimensional search spaces, where population collapse around local minima can substantially reduce global search capability.

Physics-inspired optimization methods attempt to address these issues through analogies derived from gravitational, atomic, or thermodynamic systems. Examples include the Gravitational Search Algorithm (GSA) [5], Atom Search Optimization (ASO) [6], and Atomic Orbital Search (AOS) [7]. Although such methods introduce alternative interaction mech-

anisms, many remain based on deterministic force laws or globally shared search parameters.

Recent quantum-inspired approaches have explored probabilistic search dynamics, including Quantum-Inspired Evolutionary Algorithms (QIEA) [8] and recent quantum-driven metaheuristic studies such as the Quantum Snowflake Algorithm (QSA) [9]. However, many existing quantum-inspired optimizers emphasize probabilistic representation without explicitly integrating diversity preservation and adaptive escape behavior within a single operator-level framework.

Unlike studies aiming to introduce universally dominant optimizers, the objective of QWMO is to investigate how quantum-inspired probabilistic operators influence the exploration–exploitation balance in multimodal optimization landscapes.

The main contributions of this paper are summarized as follows:

- A wave-function-inspired adaptive orbital sampling mechanism is proposed to control Gaussian search dispersion according to relative solution quality.
- A Pauli-Inspired Exclusion operator is introduced to preserve population diversity through orthogonal displacement rather than collinear repulsion.
- An Adaptive Quantum Escape operator is formulated to recover stagnating agents through probabilistic stochastic

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relocation.

- A probabilistic convergence perspective is developed using an extended-state Markov process interpretation.
- Comparative benchmark experiments and ablation analysis are conducted on representative CEC-style multimodal optimization functions.

The rest of the paper is organized as follows. Section 2 reviews related metaheuristic, physics-inspired, and quantum-inspired optimization methods. Section 3 presents the proposed QWMO framework. Section 4 discusses the probabilistic convergence perspective. Section 5 describes the experimental protocol. Section 6 reports the benchmark results. Section 7 presents the ablation study. Section 8 concludes the paper.

2. Background and Related Work

2.1. Classical population-based metaheuristics

Population-based metaheuristics maintain multiple candidate solutions and update them through stochastic or rule-based interactions. PSO models social learning through velocity-based updates [1], while GA relies on selection, crossover, and mutation to recombine candidate solutions [2]. GWO and WOA introduce biologically inspired encircling and hunting mechanisms [3, 4].

These algorithms are effective on many benchmark problems, but their performance depends strongly on the structure of the objective landscape. In particular, multimodal functions can cause premature convergence when the population loses diversity or when update rules repeatedly pull agents toward sub-optimal basins.

2.2. Physics-inspired optimization

Physics-inspired optimizers translate physical interaction models into search operators. GSA uses gravitational attraction between candidate solutions [5]. ASO models agent interactions through atom-like forces and Lennard-Jones inspired pairwise dynamics [6]. AOS uses an atomic orbital analogy to guide agent placement through probability density functions [7].

Although these approaches enrich the search process with physical metaphors, they also introduce limitations. Pairwise force models can be computationally expensive, and collinear interaction dynamics may lead to oscillatory or redundant movement. In addition, globally decaying search radii may reduce the ability to differentiate between agents that should explore and agents that should exploit.

2.3. Quantum-inspired optimization

Quantum-inspired metaheuristics use concepts such as superposition, probabilistic representation, and tunneling as algorithmic analogies rather than as direct physical simulation. QIEA is a representative example, encoding solutions through quantum-inspired probabilistic structures [8]. Recent work such as QSA

further explores quantum-driven operators in continuous and discrete optimization settings [9].

QWMO follows this broad quantum-inspired direction, but differs by explicitly decomposing the search process into three separable operators: sampling, exclusion, and escape. This decomposition enables ablation analysis and clearer assessment of the contribution of each mechanism.

2.4. Diversity preservation

Diversity preservation is a central issue in evolutionary and swarm optimization. Exclusion mechanisms, multiswarms, niching, and fitness sharing have been used to prevent premature population collapse [10, 11]. Repulsive PSO variants also use attraction and repulsion to regulate population spread [12].

Most repulsive or exclusion-based mechanisms operate along the collision vector between two agents. QWMO instead applies orthogonal displacement, moving the weaker agent into a direction perpendicular to the collision vector. This design reduces repeated back-and-forth displacement along the same axis and encourages spatial diversification.

3. Quantum Wave-function Metaheuristic Optimizer

Let the search domain be a bounded hyper-rectangle $\Omega = [\ell, u]^D$, where D is the dimensionality. The population at iteration t is denoted by $X(t) = \{\mathbf{x}_1(t), \dots, \mathbf{x}_N(t)\}$. Each agent $\mathbf{x}_i(t) \in \Omega$ has an associated objective value $f(\mathbf{x}_i)$. The best-so-far solution is denoted by $\mathbf{x}^*$.

QWMO applies three operators at each iteration: Adaptive Orbital Sampling, Pauli-Inspired Exclusion, and Adaptive Quantum Escape. The terminology is used as an optimization analogy rather than a direct physical simulation of quantum systems.

3.1. Adaptive Orbital Sampling

Each agent samples a candidate position from a Gaussian distribution centered at its current location:

$$\mathbf{x}'_i \sim \mathcal{N}(\mathbf{x}_i, \sigma_i^2(t)\mathbf{I}). \quad (1)$$

The relative quality of agent i is defined as

$$q_i = \frac{f_{\text{worst}} - f(\mathbf{x}_i)}{f_{\text{worst}} - f_{\text{best}} + \epsilon}, \quad (2)$$

where $q_i \in [0, 1]$, f_{best} is the best value in the current population, f_{worst} is the worst value, and ϵ is a small numerical stabilizer.

The maximum orbital radius and decay coefficient are given by

$$\sigma_{\text{max},i} = \gamma(u - \ell)(2 - q_i), \quad c_i = c_{\text{base}}q_i. \quad (3)$$

The sampling radius is then computed as

$$\sigma_i(t) = \max \left\{ \sigma_{\text{max},i} \exp \left(-c_i \frac{t}{T_{\text{max}}} \right), \sigma_{\text{l}} \right\}, \quad (4)$$

where $\sigma_{\text{l}} = 10^{-8}$ ensures a strictly positive sampling variance.

High-quality agents contract more rapidly and perform local refinement, whereas low-quality agents retain wider orbits and continue exploring broader regions of the search space.

3.2. Pauli-Inspired Exclusion

If two agents violate the exclusion radius,

$$\|\mathbf{x}_i - \mathbf{x}_j\| < \varepsilon, \quad (5)$$

the weaker agent is displaced orthogonally relative to the collision vector.

Let $\mathbf{x}_b$ denote the weaker agent and $\mathbf{x}_g$ the better agent. Define

$$\mathbf{v} = \mathbf{x}_b - \mathbf{x}_g. \quad (6)$$

A random vector $\mathbf{r} \sim \mathcal{N}(0, \mathbf{I})$ is projected onto the subspace orthogonal to $\mathbf{v}$:

$$\mathbf{p} = \mathbf{r} - \frac{\mathbf{r}^\top \mathbf{v}}{\|\mathbf{v}\|^2 + 10^{-10}} \mathbf{v}. \quad (7)$$

The weaker agent is updated as

$$\mathbf{x}_b \leftarrow \text{clip}\left(\mathbf{x}_b + \alpha \frac{\mathbf{p}}{\|\mathbf{p}\| + 10^{-10}}, \ell, u\right), \quad (8)$$

where

$$\alpha \sim \mathcal{U}(0.5\varepsilon, 1.5\varepsilon). \quad (9)$$

Collision detection is implemented using a k-d tree [13], reducing neighborhood search complexity from a naive $O(N^2)$ pairwise scan to approximately $O(N \log N)$ in low and moderate dimensions.

3.3. Adaptive Quantum Escape

Each agent maintains a stagnation counter k_i . When $k_i > k_s$, the agent becomes eligible for probabilistic escape:

$$P_{\text{esc}}(t) = \exp(-2\kappa(t)L_{\text{norm}}), \quad (10)$$

where

$$\kappa(t) = \kappa_0 \frac{t}{T_{\text{max}}}, \quad L_{\text{norm}} = \frac{\|\mathbf{x}_i - \mathbf{x}^*\|}{L_{\text{max}}}. \quad (11)$$

The destination is generated through a hybrid stochastic rule:

$$\mathbf{x}'_i = \beta(\mathbf{x}^* + \boldsymbol{\eta}) + (1 - \beta)\mathbf{x}_r, \quad (12)$$

where

$$\beta \sim \mathcal{U}[0, 1], \quad \boldsymbol{\eta} \sim \mathcal{N}(0, \sigma_{\text{noise}}^2 \mathbf{I}), \quad \mathbf{x}_r \sim \mathcal{U}[\ell, u]^D.$$

The global-best component provides directional guidance, while the random component preserves long-range exploration.

3.4. Algorithmic procedure

4. Theoretical Analysis

The probabilistic dynamics of QWMO can be interpreted within an extended-state Markov process. Let $\mathcal{S} \subset \mathbb{R}^D$ denote the bounded search space and let $P(\mathbf{x}, A)$ denote the stochastic transition kernel induced by the QWMO update operators for a measurable set $A \subseteq \mathcal{S}$.

Since the Adaptive Quantum Escape operator depends on the stagnation counter, the state is augmented as

$$s_i(t) = (\mathbf{x}_i(t), k_i(t)). \quad (13)$$

This extended representation preserves the Markov property because future transitions depend on the current position and current stagnation counter, not on the full historical trajectory.

Algorithm 1 QWMO

Require: $N, T_{\text{max}}, \varepsilon_r, k_s, \gamma, c_{\text{base}}, \kappa_0, \eta_r$

Ensure: Best-so-far solution $\mathbf{x}^*$

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1: Initialize  $X$  uniformly in  $[\ell, u]^D$ 
2: Evaluate  $f(\mathbf{x}_i)$  for all agents
3:  $\mathbf{x}^* \leftarrow \arg \min_i f(\mathbf{x}_i)$ 
4:  $k_i \leftarrow 0$  for all agents
5:  $\varepsilon \leftarrow \varepsilon_r(u - \ell)$ 
6:  $\sigma_{\text{noise}} \leftarrow \eta_r(u - \ell)$ 
7:  $L_{\text{max}} \leftarrow \sqrt{D}(u - \ell)$ 
8: for  $t = 1$  to  $T_{\text{max}}$  do
9:   Compute  $f_{\text{best}}$  and  $f_{\text{worst}}$ 
10:  for each agent  $i$  do
11:    Compute  $q_i, \sigma_{\text{max},i}, c_i$ , and  $\sigma_i(t)$ 
12:    Sample  $\mathbf{x}'_i \sim \mathcal{N}(\mathbf{x}_i, \sigma_i^2(t)\mathbf{I})$ 
13:    Clip  $\mathbf{x}'_i$  to  $[\ell, u]^D$ 
14:    if  $f(\mathbf{x}'_i) < f(\mathbf{x}_i)$  then
15:       $\mathbf{x}_i \leftarrow \mathbf{x}'_i$ 
16:       $k_i \leftarrow 0$ 
17:    else
18:       $k_i \leftarrow k_i + 1$ 
19:    end if
20:  end for
21:  Build k-d tree and query pairs within  $\varepsilon$ 
22:  for each colliding pair do
23:    Apply Pauli-Inspired Exclusion to the weaker agent
24:  end for
25:  for each agent with  $k_i > k_s$  do
26:    Compute  $P_{\text{esc}}(t)$ 
27:    if  $\mathcal{U}(0, 1) < P_{\text{esc}}(t)$  then
28:      Apply Adaptive Quantum Escape
29:       $k_i \leftarrow 0$ 
30:    end if
31:  end for
32:  Update  $\mathbf{x}^*$  if a better solution is found
33: end for
34: return  $\mathbf{x}^*$ 

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Lemma 1 (Positive reachability). *If $\sigma_i(t) \geq \sigma_l > 0$, then the Adaptive Orbital Sampling operator assigns non-zero probability to every measurable neighborhood within the bounded search domain.*

Proof. The truncated Gaussian sampling kernel has strictly positive density over the bounded search domain when its variance is bounded below by $\sigma_l > 0$. Therefore, for any measurable set A with positive Lebesgue measure, $P(\mathbf{x}, A) > 0$. $\square$

Lemma 2 (Stagnation escape). *The Adaptive Quantum Escape operator prevents permanent absorbing stagnation states under the idealized stochastic model.*

Proof. Whenever an agent fails to improve, its stagnation counter increases. Once the counter exceeds k_s , the escape operator is activated with strictly positive probability as long as $\kappa(t)$ and L_{norm} are finite. Hence, stagnant states retain non-zero probability of leaving their local region. $\square$

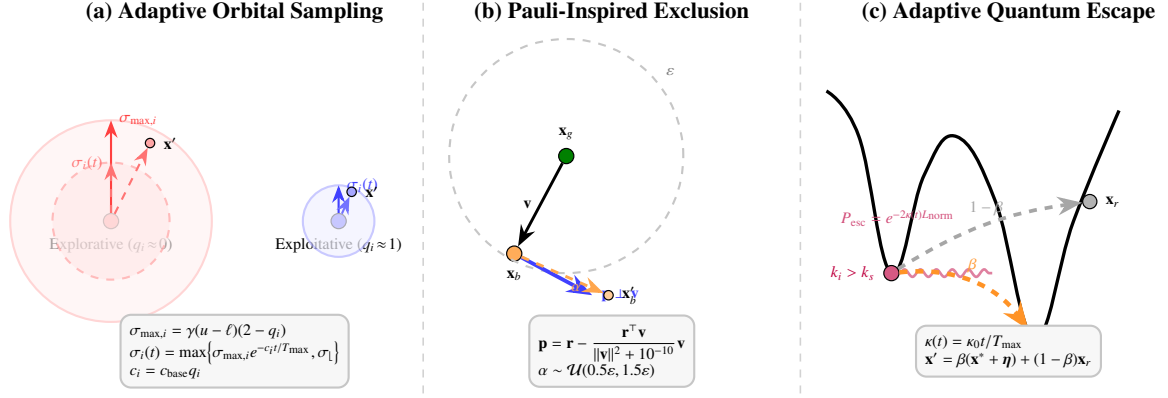


Figure 1: Schematic illustration of the three operators composing QWMO. (a) Adaptive Orbital Sampling dynamically adjusts the Gaussian sampling radius according to solution quality, balancing exploration and exploitation. (b) Pauli-Inspired Exclusion applies orthogonal displacement when neighboring agents violate the exclusion radius ε , promoting spatial diversity. (c) Adaptive Quantum Escape enables stagnating agents to leave local minima through probabilistic β -hybrid stochastic relocation between the global-best region and random exploration targets. The terminology is used as an optimization analogy rather than a direct physical simulation of quantum systems. The combined interaction of the three operators establishes the exploration–exploitation balance of QWMO.

Theorem 1 (Practical stochastic convergence behavior). *Under bounded-domain assumptions and strictly positive transition probabilities induced by the sampling and escape operators, QWMO exhibits practical stochastic convergence behavior toward high-quality regions of the search space.*

Remark 1. *This result is stated as a probabilistic convergence perspective rather than a full asymptotic global convergence proof. A complete Harris recurrence argument would require explicit drift and minorization conditions, which are left for future theoretical work.*

5. Experimental Setup

5.1. Benchmark functions

The proposed algorithm was evaluated on five representative CEC-style benchmark functions: Sphere (F1), Schwefel (F5), Rastrigin (F10), Hybrid Function (F20), and Composition Function (F28). These functions cover unimodal, multimodal, hybrid, and composition landscapes.

All experiments were conducted in 30 dimensions using 30 independent runs with distinct random seeds. The maximum iteration budget was fixed at

$$T_{\text{max}} = 300000.$$

5.2. Compared algorithms

QWMO was compared against six baseline algorithms: PSO, GA, GWO, WOA, ASO, and AOS. These baselines were selected to represent classical swarm intelligence, evolutionary search, and physics-inspired optimization.

5.3. Parameter settings

The QWMO parameters were fixed as follows:

$$\gamma = 0.05, \quad c_{\text{base}} = 5, \quad \kappa_0 = 8, \quad \varepsilon_r = 0.05, \quad k_s = 10, \quad \eta_r = 0.001$$

The population size was fixed at $N = 50$ for all algorithms.

5.4. Statistical analysis

Performance is reported as mean fitness $\pm$ standard deviation over 30 independent runs. Pairwise statistical significance between QWMO and each competing algorithm was evaluated using Wilcoxon signed-rank analysis with significance threshold $p < 0.05$.

Pairwise Wilcoxon signed-rank analysis was adopted as the primary statistical comparison framework in this preliminary study. No global rank-based post-hoc procedure is reported.

6. Results and Discussion

Table 1 reports the mean and standard deviation of the best fitness values obtained over 30 runs. Lower values indicate better performance.

QWMO consistently improves over the physics-inspired baselines ASO and AOS on most benchmark functions. In particular, the large performance gap on F10 Rastrigin indicates that the adaptive escape mechanism improves search behavior on highly multimodal landscapes.

At the same time, QWMO does not dominate all classical baselines. GA obtains the strongest results on several functions, and PSO/GWO achieve exact or near-exact convergence on the unimodal Sphere function. These results support a mechanism-focused interpretation: QWMO is not a universal replacement for classical optimizers, but a probabilistic framework designed to enhance exploration and escape behavior in multimodal settings.

Table 2 reports pairwise Wilcoxon comparisons between QWMO and each competitor.

The Wilcoxon results show that QWMO is statistically stronger than AOS across all five functions and stronger than ASO on four out of five functions. This supports the claim that the proposed operator combination improves over closely related physics-inspired baselines. Against mature classical optimizers, QWMO is competitive but not uniformly superior.

Table 1: Mean $\pm$ standard deviation of best fitness values over 30 independent runs in 30 dimensions. Bold values indicate the best result in each column.

Algorithm	F1 Sphere	F5 Schwefel	F10 Rastrigin	F20 Hybrid	F28 Composition
PSO	0.000e+00 $\pm$ 0.00e+00	4.619e+03 $\pm$ 7.83e+02	5.888e+01 $\pm$ 2.05e+01	1.647e+02 $\pm$ 8.06e+00	1.066e+04 $\pm$ 1.13e+02
GA	2.717e-05 $\pm$ 1.17e-05	3.158e+02 $\pm$ 2.25e+02	8.339e-06 $\pm$ 3.41e-06	1.486e+02 $\pm$ 1.42e-09	1.055e+04 $\pm$ 2.78e-07
GWO	0.000e+00 $\pm$ 0.00e+00	9.255e+03 $\pm$ 2.62e+02	0.000e+00 $\pm$ 0.00e+00	1.486e+02 $\pm$ 1.14e-02	1.112e+04 $\pm$ 5.26e+01
WOA	0.000e+00 $\pm$ 0.00e+00	4.038e+03 $\pm$ 9.52e+02	1.587e+02 $\pm$ 4.51e+01	1.817e+02 $\pm$ 2.02e+01	1.055e+04 $\pm$ 5.97e-10
ASO	6.668e+02 $\pm$ 2.49e+03	3.871e+03 $\pm$ 6.83e+02	1.308e+02 $\pm$ 2.69e+01	1.805e+02 $\pm$ 1.12e+01	1.059e+04 $\pm$ 6.83e+01
AOS	3.622e+03 $\pm$ 5.39e+02	7.258e+03 $\pm$ 2.75e+02	1.877e+02 $\pm$ 1.48e+01	2.074e+02 $\pm$ 6.96e+00	1.060e+04 $\pm$ 8.96e+00
QWMO	1.082e+00 $\pm$ 2.65e-01	3.028e+03 $\pm$ 4.23e+02	9.023e+00 $\pm$ 2.13e+00	1.506e+02 $\pm$ 6.95e-01	1.059e+04 $\pm$ 2.67e+01

Table 2: Wilcoxon signed-rank test results with QWMO as reference. Symbol + indicates QWMO is significantly better; – indicates the competitor is significantly better; $\approx$ indicates no statistically significant difference at $p < 0.05$.

vs. QWMO	F1	F5	F10	F20	F28
PSO	–	+	+	+	+
GA	–	–	–	–	–
GWO	–	+	–	–	+
WOA	–	+	+	+	–
ASO	+	+	+	+	$\approx$
AOS	+	+	+	+	+

7. Ablation Study

To isolate the contribution of each operator, three nested configurations were evaluated:

- **Orbital Only:** Adaptive Orbital Sampling alone.
- **Orbital + Pauli:** Adaptive Orbital Sampling and Pauli-Inspired Exclusion.
- **Full QWMO:** all three operators enabled.

The experimental protocol mirrors the main benchmark: 30 independent runs in 30 dimensions.

The Orbital Only configuration provides a strong local exploitation baseline but stagnates on multimodal landscapes. Adding Pauli-Inspired Exclusion does not produce a measurable improvement at $D = 30$. This behavior is explained by geometric sparsity: with $\varepsilon_r = 0.05$ and a function range approximately spanning $[-100, 100]^D$, the exclusion radius is $\varepsilon = 0.05 \times 200 = 10$, while the search-space diameter is

$$L_{\max} \approx \sqrt{30} \times 200 \approx 1095.$$

Thus, agents distributed across a high-dimensional domain rarely fall within the exclusion radius. This confirms that the Pauli operator requires an adaptive radius $\varepsilon(t)$ to be more effective in high-dimensional search spaces.

The Full QWMO configuration substantially improves performance on multimodal and hybrid functions. On F10 Rastrigin, the mean error decreases from 52.20 to 1.40, while on F5 Schwefel it decreases from approximately 3126 to 2088. These results indicate that Adaptive Quantum Escape is the primary contributor to performance gains on deceptive landscapes.

However, the Full QWMO configuration performs worse on Sphere and does not improve the Composition function. This trade-off is consistent with an exploration-oriented design: stochastic escape improves recovery from local minima, but can reduce fine precision on smooth unimodal surfaces or fail to resolve highly structured composition landscapes.

8. Conclusion

This paper presented QWMO, a quantum wave-function inspired metaheuristic framework for multimodal optimization. The algorithm combines Adaptive Orbital Sampling, Pauli-Inspired Exclusion, and Adaptive Quantum Escape within a unified probabilistic search process.

The experimental results show that QWMO consistently improves over its direct physics-inspired counterparts ASO and AOS, while remaining competitive with established swarm-based optimizers on multimodal and hybrid functions. The ablation study indicates that the strength of QWMO does not arise from a single operator, but from the interaction between adaptive orbital sampling, diversity-preserving exclusion, and stochastic escape dynamics.

Several limitations remain. The fixed exclusion radius is not sufficiently effective in sparse high-dimensional populations, and the current escape policy does not materially improve the hardest composition landscape. Future work will investigate adaptive exclusion radii, larger benchmark suites, high-dimensional scalability, and stronger stochastic convergence guarantees based on Harris recurrence theory.

Table 3: Ablation study results for QWMO variants (mean $\pm$ standard deviation, 30D, 30 runs). Bold values indicate the best ablation configuration.

Configuration	F1 Sphere	F5 Schwefel	F10 Rastrigin	F20 Hybrid	F28 Composition
Orbital Only	5.45e-02 $\pm$ 3.62e-03	3.13e+03 $\pm$ 2.08e+02	5.22e+01 $\pm$ 7.54e+00	1.61e+02 $\pm$ 2.34e+00	1.055e+04 $\pm$ 1.62e-03
Orbital + Pauli	8.08e-02 $\pm$ 7.71e-03	3.13e+03 $\pm$ 2.08e+02	5.22e+01 $\pm$ 7.54e+00	1.61e+02 $\pm$ 2.34e+00	1.055e+04 $\pm$ 3.71e-03
Full QWMO	5.61e-01 $\pm$ 1.06e-01	2.09e+03 $\pm$ 3.28e+02	1.40e+00 $\pm$ 9.02e-01	1.50e+02 $\pm$ 6.56e-02	1.055e+04 $\pm$ 3.91e-02

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References

- [1] J. Kennedy, R. Eberhart, Particle swarm optimization, in: Proceedings of ICNN'95 – International Conference on Neural Networks, Vol. 4, IEEE, 1995, pp. 1942–1948. doi:10.1109/ICNN.1995.488968.
- [2] D. E. Goldberg, Genetic Algorithms in Search, Optimization and Machine Learning, Addison-Wesley, Reading, MA, 1989.
- [3] S. Mirjalili, S. M. Mirjalili, A. Lewis, Grey wolf optimizer, Advances in Engineering Software 69 (2014) 46–61. doi:10.1016/j.advengsoft.2013.12.007.
- [4] S. Mirjalili, A. Lewis, The whale optimization algorithm, Advances in Engineering Software 95 (2016) 51–67. doi:10.1016/j.advengsoft.2016.01.008.
- [5] E. Rashedi, H. Nezamabadi-Pour, S. Saryazdi, GSA: A gravitational search algorithm, Information Sciences 179 (13) (2009) 2232–2248. doi:10.1016/j.ins.2009.03.004.
- [6] W. Zhao, L. Wang, Z. Zhang, Atom search optimization and its application to solve a hydrogeological parameter estimation problem, Knowledge-Based Systems 163 (2019) 283–304. doi:10.1016/j.knosys.2018.08.030.
- [7] M. Azizi, Atomic orbital search: A novel metaheuristic algorithm, Applied Mathematical Modelling 93 (2021) 657–683. doi:10.1016/j.apm.2020.12.021.
- [8] R. Nowotniak, J. Kucharski, Survey of quantum-inspired evolutionary algorithms, in: Proceedings of the Federated Conference on Computer Science and Information Systems, 2010, pp. 455–462.
- [9] Z. Oralhan, B. Oralhan, Quantum snowflake algorithm (QSA): A snowflake-inspired quantum-driven metaheuristic for continuous and discrete optimization, Applied Sciences 15 (9) (2025) 5117. doi:10.3390/app15095117.
- [10] T. Blackwell, J. Branke, Multiswarms, exclusion, and anti-convergence in dynamic environments, IEEE Transactions on Evolutionary Computation 10 (4) (2006) 459–472. doi:10.1109/TEVC.2005.857074.
- [11] B. Sareni, L. Krähenbühl, Fitness sharing and niching methods revisited, IEEE Transactions on Evolutionary Computation 2 (3) (1998) 97–106. doi:10.1109/4235.735432.
- [12] R. J. Mullen, D. Monekosso, S. Barman, P. Remagnino, Solving the orienteering problem using attractive and repulsive particle swarm optimization, in: Proceedings of the IEEE Congress on Evolutionary Computation, 2007, pp. 2352–2359. doi:10.1109/CEC.2007.4424765.
- [13] J. L. Bentley, Multidimensional binary search trees used for associative searching, Communications of the ACM 18 (9) (1975) 509–517. doi:10.1145/361002.361007.